

1) List the characteristics of IoT.

Characteristics of IoT

1. Interconnectivity – Devices are connected through the internet.
 2. Intelligence – Smart devices can process and analyze data.
 3. Dynamic nature – Device states change continuously.
 4. Scalability – Supports large number of connected devices.
 5. Heterogeneity – Different devices and platforms work together.
 6. Self-configuration – Devices configure automatically.
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2) Define Arduino Interrupt.

Arduino Interrupt

An interrupt is a signal that temporarily stops the normal execution of a program and executes a special function called Interrupt Service Routine (ISR).

It helps Arduino respond immediately to external events like button press or sensor signal.

3) What is the requirement of IoT protocol standardization?

Requirement of IoT Protocol Standardization

1. Ensures interoperability between different IoT devices.
 2. Provides secure and reliable communication.
 3. Reduces complexity in device integration.
 4. Enables scalability and compatibility across networks.
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4) List out the important features of Wi-Fi.

Features of Wi-Fi

1. Wireless communication technology based on IEEE 802.11.
 2. Provides high-speed internet access.
 3. Supports mobility and portability.
 4. Easy installation and low maintenance.
 5. Supports infrastructure and ad-hoc modes.
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5) State where Zigbee makes its mark with regard to IoT applications.

Applications of Zigbee in IoT

1. Home automation systems
 2. Smart lighting systems
 3. Industrial monitoring
 4. Security and surveillance systems
 5. Smart energy management
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6) Differentiate between SPI and I2C protocol in Raspberry Pi.

SPI	I2C
Uses 4 wires	Uses 2 wires
High speed communication	Comparatively slower
Full duplex communication	Half duplex communication
No addressing required	Uses device addressing
Suitable for short distance	Suitable for multiple devices

7) Define Smart Irrigation System

A Smart Irrigation System is an IoT-based automated watering system used in agriculture. It uses sensors such as soil moisture, temperature, and humidity sensors to monitor field conditions. The system automatically supplies water only when required, reducing water wastage and improving crop yield.

Advantages

- Saves water and power
 - Reduces human effort
 - Improves agricultural productivity
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8) What is Industrial Automation?

Industrial Automation is the use of control systems, computers, sensors, robots, and machines to perform industrial operations automatically with minimum human intervention. It increases production efficiency, accuracy, and safety in industries.

Benefits

- Reduces manual labor
 - Improves productivity
 - Increases operational safety
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9) What type of semiconductor is used for Gas Detection?

Metal Oxide Semiconductor (MOS) is commonly used for gas detection in embedded and IoT systems. The resistance of the semiconductor changes when exposed to gases, helping in gas sensing and detection.

Examples

- MQ-2 Gas Sensor
- MQ-135 Air Quality Sensor

Applications

- Gas leakage detection
 - Air pollution monitoring
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10) List some current smart wearable devices.

Smart wearable devices are electronic devices worn on the body to monitor health, fitness, and daily activities.

Examples

1. Smart watches
 2. Fitness bands
 3. Smart glasses
 4. Smart shoes
 5. Health monitoring bands
 6. Smart rings
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11(a) Discuss about the Zigbee architecture with neat diagram.

Zigbee Architecture

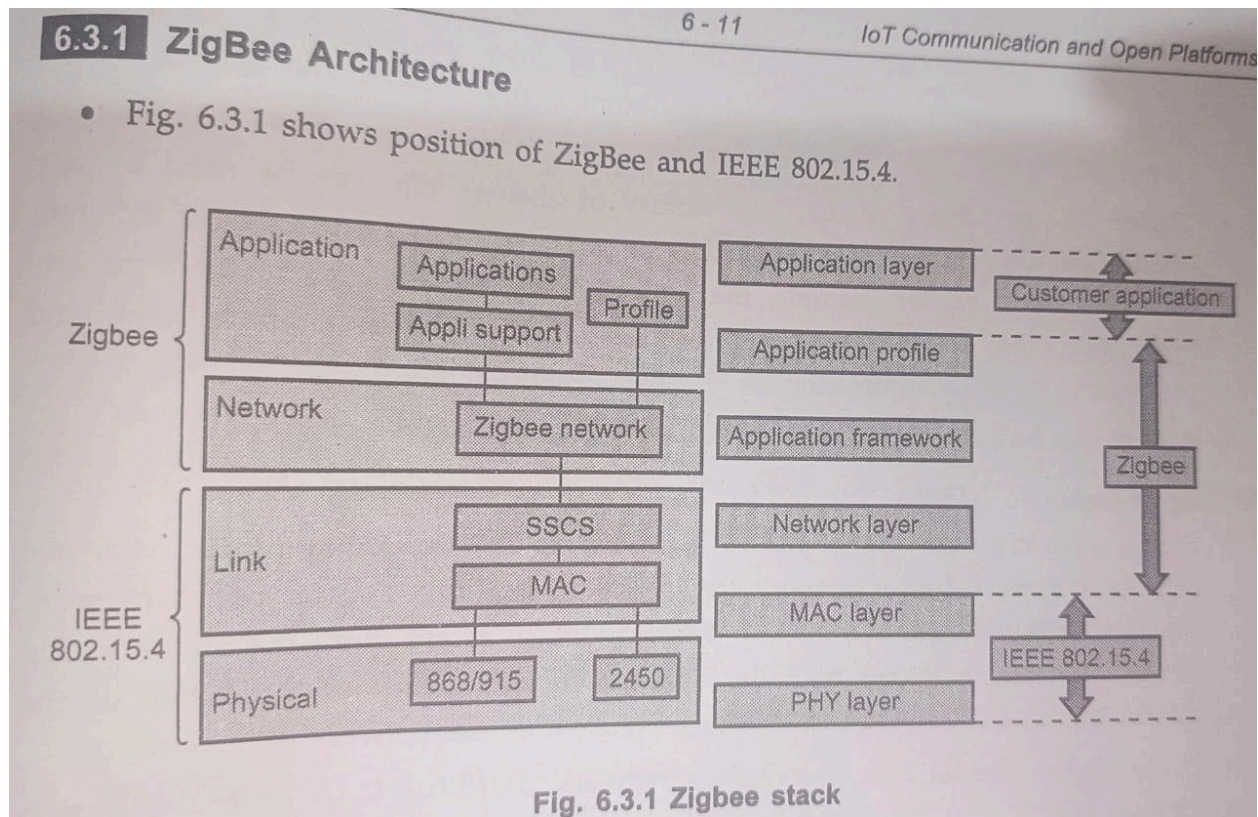
Zigbee is a low-power, low-cost wireless communication technology based on the IEEE 802.15.4 standard. It is mainly used for IoT applications such as home automation, industrial monitoring, smart lighting, security systems, and wireless sensor networks.

Features

- Low power consumption

- Low cost
- Supports mesh networking
- Data rate up to 250 kbps
- Suitable for IoT applications

Zigbee Architecture Layers



1. Physical Layer

The Physical layer is the lowest layer of Zigbee architecture. It is responsible for transmitting and receiving data signals.

Functions

- Channel selection
- Data transmission and reception
- Modulation and demodulation
- Energy detection

Features

- Operates in 2.4 GHz ISM band

- Supports data rates of 20 kbps, 40 kbps, 100 kbps, and 250 kbps
- Uses low transmission power

Advantages

- Low power usage
 - Reliable wireless communication
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2. MAC Layer

The MAC layer is defined by IEEE 802.15.4 standard. It controls how devices access the communication channel.

Functions

- Channel access using CSMA/CA
- Beacon frame management
- Error detection and correction
- Acknowledgement handling
- Synchronization between devices

Important Features

- Supports guaranteed time slots (GTS)
 - Provides reliable communication
 - Reduces collision between devices
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3. Network Layer

The Network layer manages network formation and routing operations.

Functions

- Device joining and leaving
- Route discovery and routing
- Network configuration
- Security management

Advantages

- Provides scalable networking
 - Enables long-distance communication through routing
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4. Application Layer

The Application layer is the top layer of Zigbee architecture. It provides communication between user applications and Zigbee devices.

Functions

- Device discovery
- Service management
- Data transmission to applications
- Security and authentication

Applications

- Smart home devices
 - Industrial IoT systems
 - Healthcare monitoring systems
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Types of Devices in Zigbee

1. Full Function Device (FFD)

- Acts as coordinator or router.
- Can communicate with all devices.

2. Reduced Function Device (RFD)

- Simple end device with low memory.
 - Works under control of FFD.
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Zigbee Topologies

1. Star Topology

All devices communicate through coordinator.

2. Tree Topology

Devices arranged in hierarchical structure.

3. Mesh Topology

Devices communicate through multiple paths.

Advantages of Zigbee

1. Very low power consumption
 2. Low cost implementation
 3. Supports large number of devices
 4. Reliable communication
 5. Secure data transfer
 6. Long battery life
 7. Easy installation and maintenance
-

Disadvantages of Zigbee

1. Low data rate
 2. Limited communication range
 3. Not suitable for high bandwidth applications
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Applications of Zigbee

1. Home automation systems
 2. Smart lighting systems
 3. Industrial automation
 4. Healthcare monitoring
 5. Security and surveillance systems
 6. Smart agriculture
 7. Wireless sensor networks
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12 (a) Illustrate the IoT technology impacting the healthcare sector in fitness monitoring and wearable electronics. (16 Marks)

IoT in Healthcare – Fitness Monitoring and Wearable Electronics

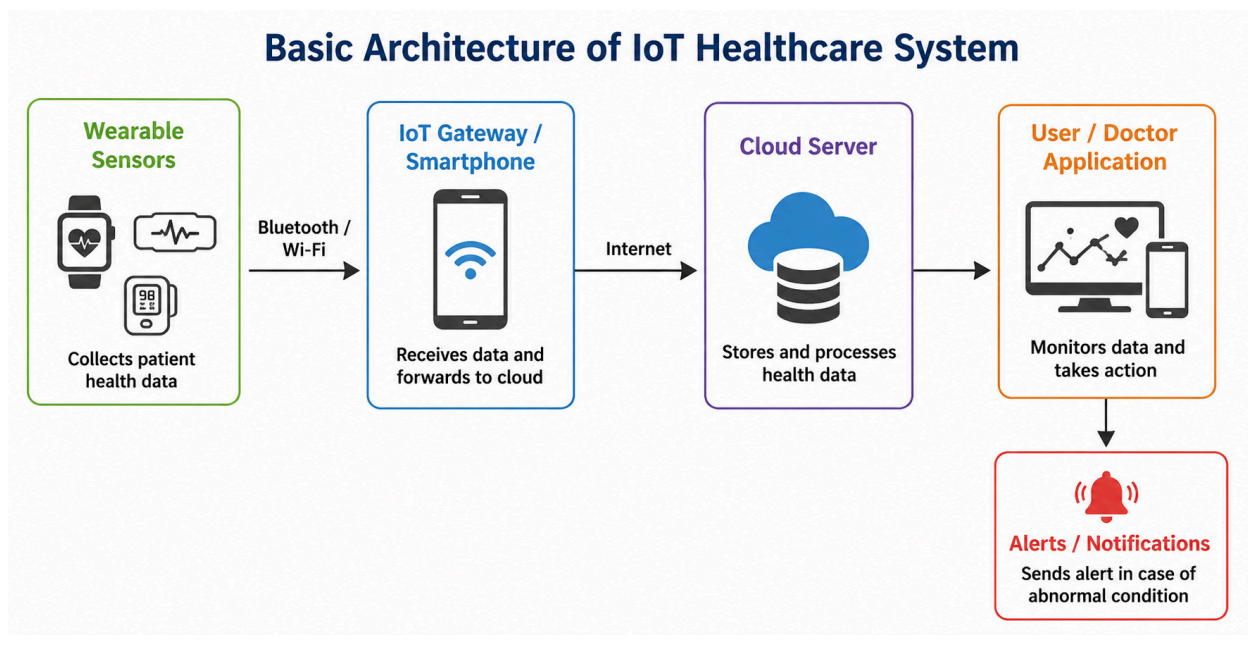
The Internet of Things (IoT) plays an important role in modern healthcare systems. IoT connects smart medical devices, sensors, wearable electronics, and cloud platforms through the internet to monitor patient health in real time.

Fitness monitoring and wearable electronics help doctors and users continuously track health conditions and improve healthcare services.

Definition

IoT in healthcare refers to the use of interconnected smart devices and sensors to collect, transmit, and analyze medical and fitness data through the internet.

Basic Architecture of IoT Healthcare System



Working Principle

1. Wearable sensors collect health data from the human body.
 2. The collected data is sent to smartphones or IoT gateways using Bluetooth or Wi-Fi.
 3. The gateway uploads data to cloud servers through the internet.
 4. Doctors or users monitor health conditions through mobile applications or dashboards.
 5. Alerts are generated during emergency situations
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Wearable Electronics in Healthcare

Wearable electronics are smart electronic devices worn on the body to monitor fitness and medical conditions.

Types of Wearable Devices

1) Smart Watches

Smart watches are wearable electronic devices worn on the wrist. They monitor heart rate, steps, sleep patterns, and fitness activities. They also provide notifications and internet connectivity.

Examples: Apple Watch, Samsung Galaxy Watch

2) Fitness Bands

Fitness bands are lightweight wearable devices mainly used for fitness tracking. They monitor physical activities such as walking, calories burned, pulse rate, and exercise performance.

Examples: Mi Band, Fitbit

3) Smart Glasses

Smart glasses are wearable devices with built-in sensors and display technology. They provide health information, navigation, and augmented reality features.

Applications: Medical assistance and real-time information display

4) Smart Clothing

Smart clothing contains embedded sensors and electronic components that monitor body temperature, heart rate, and body movements.

Applications: Sports monitoring and healthcare monitoring

5) Wearable ECG Monitors

Wearable ECG monitors continuously measure heart activity and detect abnormal heart conditions. They are mainly used for cardiac patient monitoring.

Applications: Heart disease monitoring and emergency alerts

6) Smart Shoes

Smart shoes contain sensors that track walking patterns, distance traveled, posture, and physical activities.

Applications: Fitness tracking and sports analysis

7) Health Monitoring Bands

Health monitoring bands monitor vital health parameters such as oxygen level, blood pressure, and pulse rate in real time.

Applications: Patient monitoring and elderly care systems

Sensor	Purpose
Heart Rate Sensor	Measures pulse rate
Temperature Sensor	Measures body temperature
ECG Sensor	Monitors heart activity
Blood Pressure Sensor	Measures BP
SpO2 Sensor	Measures oxygen level
Motion Sensor	Detects movement and activity

Applications of IoT in Healthcare

1) Remote Patient Monitoring

IoT devices continuously monitor patient health conditions from remote locations. Doctors can access real-time health data through cloud platforms.

Applications

- Monitoring heart patients
 - Diabetes monitoring
 - Home healthcare systems
-

2) Fitness Tracking

Wearable devices help users monitor daily physical activities and fitness levels.

Applications

- Step counting
 - Calorie tracking
 - Sleep monitoring
 - Exercise monitoring
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3) Elderly Care

IoT devices provide safety and health monitoring for old-age people.

Applications

- Fall detection systems
 - Emergency alert systems
 - Health tracking for senior citizens
-

4) Chronic Disease Management

IoT helps monitor patients suffering from long-term diseases continuously.

Applications

- Blood pressure monitoring
 - ECG monitoring
 - Asthma monitoring systems
-

5) Smart Hospitals

IoT technology automates hospital operations and patient monitoring systems.

Applications

- Smart ICU monitoring
 - Smart medical equipment tracking
 - Automated patient records
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6) Emergency Alert Systems

IoT systems automatically send alerts during medical emergencies.

Applications

- Accident detection
- Emergency ambulance alerts

- Critical patient notifications
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Advantages of IoT in Healthcare

- Real-time health monitoring
 - Early disease detection
 - Reduced hospital visits
 - Improved patient care
 - Remote patient monitoring
 - Faster emergency response
 - Better fitness management
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Challenges in IoT Healthcare

1. Data privacy and security issues
 2. Internet dependency
 3. High implementation cost
 4. Device compatibility problems
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Real-Time Example

A fitness band measures heart rate and oxygen level continuously. The data is sent to a smartphone through Bluetooth and uploaded to cloud storage. If abnormal conditions occur, alert messages are sent to doctors or family members immediately.

Future Scope of IoT in Healthcare

1. AI-based health prediction systems
 2. Smart robotic surgery
 3. Advanced wearable medical devices
 4. Smart ambulances with IoT monitoring
 5. Cloud-based healthcare management systems
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13) Explain the features of Arduino and discuss types of Arduino boards with sketch

and pin structures. (8 Marks)

Arduino

Arduino is an open-source microcontroller platform used for embedded systems and IoT applications. It is easy to program and interface with sensors, actuators, and communication modules.

Features of Arduino

1. Low-cost and easy-to-use platform
 2. Open-source hardware and software
 3. Simple programming using Arduino IDE
 4. Supports digital and analog input/output pins
 5. USB interface for programming and communication
 6. Supports sensors, motors, and IoT modules
 7. Compact size with low power consumption
 8. Supports multiple communication protocols like UART, SPI, and I2C
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Types of Arduino Boards

1) Arduino UNO

- Most commonly used Arduino board
 - Based on ATmega328P microcontroller
 - Suitable for beginners and IoT projects
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2) Arduino Mega

- Provides more memory and GPIO pins
 - Suitable for large embedded projects
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3) Arduino Nano

- Small-sized and compact board
 - Breadboard friendly and portable
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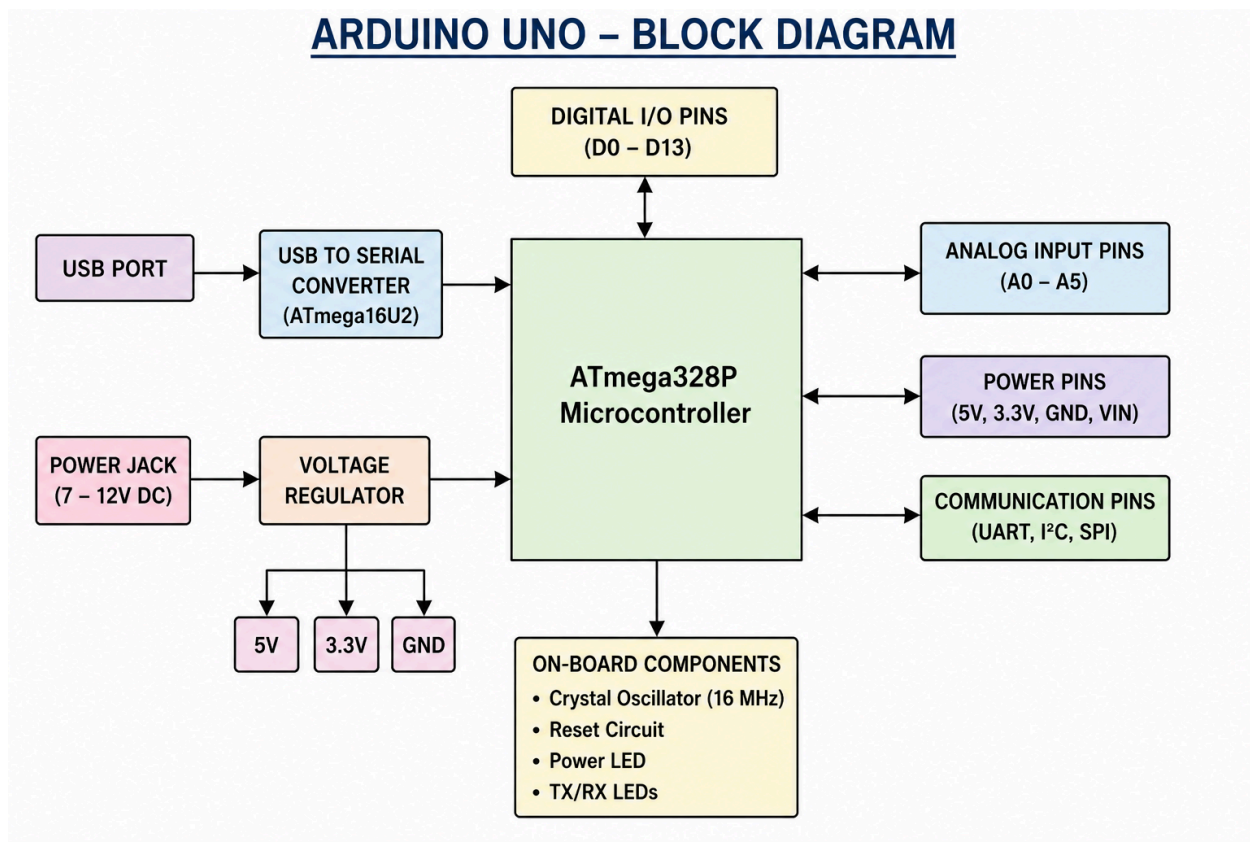
4) Arduino Leonardo

- Supports direct USB communication
 - Used for keyboard and mouse emulation projects
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5) Arduino Due

- High-speed ARM Cortex processor board
 - Used for advanced applications
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Arduino UNO Sketch



Pin Structure of Arduino UNO

Digital Pins

- Pins 0 to 13 are digital input/output pins.

- Used to connect LEDs, switches, and sensors.
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Analog Pins

- Pins A0 to A5 are analog input pins.
 - Used to read analog sensor values.
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PWM Pins

- Used for Pulse Width Modulation operations.
 - Helps control motor speed and LED brightness.
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Power Pins

- Includes 3.3V, 5V, GND, and Vin pins.
 - Provides power supply to external devices.
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Communication Pins

- UART, SPI, and I2C pins used for serial communication.
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Applications of Arduino

- Home automation systems
- Robotics projects
- Smart irrigation systems
- Healthcare monitoring systems
- Industrial automation
- IoT-based sensor applications